The Topology of Market Regimes

An Expository Research Paper on the Beta-to-Janus Framework

Adapted from Karim Khem's lesson
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Abstract

Today we will talk about the topology of market regimes and the distinction between a moving market state and a changing classification framework. Following Karim Khem's Beta-to-Janus construction, this paper develops a two-dimensional variance state space, normalizes each leg's variance against its historical reference, and compares fragile density with antifragile pressure. The equality of these densities yields an explicit boundary whose position depends on estimated masses and variance references. Under positive-mass and positive-reference assumptions, a continuous, reversible change of coordinates straightens the boundary and shows that each strict-dominance domain is path-connected and contractible. Source-reported examples from March 2016 illustrate the reclassification of a fixed state when the framework changes; an April 2020 example illustrates the appearance of a previously absent domain. Instantaneous geometric dominance is distinguished from temporal confirmation in the trading algorithm. The paper concludes with a computational reconstruction exercise. Historical values and figures are reproduced from the archived lesson; this exposition does not introduce a new empirical backtest.

Keywords: market regimes; Janus; variance stress; boundary deformation; connectedness; contractibility.

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Scope, method, and notation

This is an expository adaptation of the channel lesson, rather than a report of new experiments. The substantive explanation is retained in the source's instructional voice and organized into numbered sections. The mathematical attachments have been transcribed into native LaTeX; the five diagrams remain the original figures. Historical parameter values are reported as given in the lesson. Repository and reel links supplied in the channel are listed as source material, without a claim that their code or data have been independently re-run for this edition.

For the regular two-domain analysis, take $X = [0, \infty)^2$ in variance coordinates and assume $b_{+,t} > 0$, $b_{-,t} > 0$, $f_t > 0$, and $n_t > 0$. Neighborhoods and boundaries are understood relative to X. The change of coordinates $(V_-, V_+) \mapsto (V_-, V_+/C_t(V_-))$ is the reversible transformation used in the topology argument; the clipped variance-to-stress map itself need not be invertible. The separately discussed case $n = 0$, $f > 0$ is outside the regular positive-mass case.

Symbol	Meaning
R_t^+, R_t^-	Weighted logarithmic returns of the two Beta legs.
V_t^+, V_t^-	Annualized rolling variances, using 63 observations in the lesson.
$b_{+,t}, b_{-,t}$	Historical variance references: medians of the previous 252 variance observations in the stated configuration.
s_+, s_-	Nonnegative variance stress relative to each reference.
f, n	Admissible fragile and antifragile response masses.
ρ_+, ρ_-, D	Two densities and their signed difference.
F_t, A_t, Γ_t	Strict-dominance domains and their equality boundary.
$C_t(V_-)$	Boundary height in variance coordinates.
G_t, τ_t	Temporally confirmed state and confirmation duration.

Source prerequisites: the Janus and Beta repositories and the Instagram reel supplied with the lesson (see References).

1 Introduction: geometry, topology, and market regimes

Source passage: Karim Khem, October 23, 2026.

In my Reel, I explained that in financial markets, it's not just the price that moves inside a framework. The framework itself can deform, and that deformation can change your market regime.

Now I'm going to show you how we can demonstrate this mathematically, using the two regimes of my Janus model: fragile and antifragile.

We're going to do this progressively. You don't need to understand all the mathematics behind the model to follow this first lesson.

I gave you the example of a drop of milk that changes shape. Now imagine taking that drop and gradually stretching it.

It becomes longer, narrower, maybe even completely flattened. But there's one thing that hasn't changed: it's still one single piece.

You can pick two points inside the drop and draw a path between them without ever leaving the drop.

Now imagine splitting that drop into two separate drops.

This time, you can no longer move from one to the other while staying inside the milk. The way those regions are connected has changed.

And that brings us to a fundamental distinction.

Geometry studies things like the shape of our drop of milk, while topology studies neighborhoods, continuity, connections, and separations, like what happens when we split that drop into two.

And that's precisely what interests me when it comes to market regimes.

I want to understand what the domain of each regime looks like, where its boundary is, whether the states inside it are connected, and what happens when market conditions change.

Let's take our first regime, which I'll call F, for fragile.

I want to gather all the states belonging to this regime into one set.

Mathematically, I can write: $F = \{\text{all states belonging to the fragile regime}\}$

The curly brackets simply mean "the set of."

So this expression tells us: "I take all the states corresponding to the fragile regime and put them together into one set."

I do exactly the same thing for the antifragile regime, which I'll call A.

$A = \{\text{all states belonging to the antifragile regime}\}$

Now ask yourself a question: how do I know whether a state belongs to F rather than A?

I need a mathematical rule that allows me to distinguish between them.

Without that rule, I can draw two regions on a chart, but I don't know why they have their particular shapes. I don't know where their boundary should be. And more importantly, I don't know how that boundary should react when conditions change.

So, to construct a topology that we can actually use in a trading model, we first need to define our domains mathematically.

And that's exactly what we're going to do with Janus.

But before that, you need to understand briefly where the data used by the model comes from.

In my **Beta and Trading Environment repository**, I construct what I call an operational beta. This isn't simply the traditional beta coefficient used to measure an asset's sensitivity to a market index.

Here, I'm working with a mathematical object made up of relationships between multiple tickers, subject to the real constraints of a trading environment.

This object includes, among other things, two sets of tickers organized into two legs, which I call R^+ and R^-.

Each ticker has a logarithmic return, the same one we discussed in the Reel:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{1}$$

2 Returns and the construction of the two Beta legs

Source passage: Karim Khem, October 23, 2026.

I take the current price of a ticker, divide it by its previous price, and then take the logarithm of the result."

The i simply identifies the ticker, and t identifies the date.

Now, inside my two legs, each ticker also has a weight. One asset might represent 10% of the basket, another 20%, and so on.

The return of a leg can therefore be written as:

$$R_t^+ = \sum_{i \in U^+} w_{i,t}^+ r_{i,t} \tag{2}$$

Look at this equation.

The symbol Σ, which might seem complicated at first, simply means "add everything together."

U^+ is the set of tickers in the first leg. w represents each ticker's weight, and r represents its return.

So the entire equation tells us:

The return of my first leg is equal to the sum of its tickers' returns, each multiplied by its weight

You can check that with a very simple example.

Imagine two tickers. The first represents 60% of the basket and has a logarithmic return of 0.02. The second represents 40% and has a logarithmic return of -0.01.

$$R^+ = 0.60 \times 0.02 + 0.40 \times (-0.01) \tag{3}$$

$$R^+ = 0.008 \tag{4}$$

See?

We've simply added two weighted contributions.

For the other leg, I apply exactly the same principle:

$$R_t^- = \sum_{j \in U^-} w_{j,t}^- r_{j,t} \tag{5}$$

The $+$ and $-$ signs identify the two legs.

They don't mean their returns must always be positive or negative.

In Beta, the weights depend partly on the assets' historical volatility, with concentration constraints. The relationships between the two baskets are also studied through their cumulative levels and residuals. You can find the detailed construction in the Beta notebook.

Why is this relevant to topology?

Because we're no longer working with a single isolated price trajectory. We have two legs that evolve, and the relationships between them can evolve as well.

So we're back to the central idea of the Reel: there's the movement of the individual elements, but there's also the evolution of the relationships that organize the system.

For this first lesson, we're going to keep the two legs separate.

I take their cumulative logarithmic returns, which I call L^+ and L^-, and transform them into two positive indices:

$$P_t^{\pm} = \exp\Big(L_t^{\pm} - L_{t_0}^{\pm}\Big) \tag{6}$$

I take the cumulative evolution of each leg from the starting date and transform it into a positive index that begins at 1.

Why do this?

Because the Janus operator works with two positive series and calculates their logarithmic returns.

And we can demonstrate that this transformation preserves the returns of the two legs:

$$\ln\left(\frac{P_t^{\pm}}{P_{t-1}^{\pm}}\right) = R_t^{\pm} \tag{7}$$

3 Variance, historical references, and stress

Source passage: Karim Khem, October 23, 2026.

When Janus calculates the logarithmic return of these new indices, it recovers the original return of each Beta leg

That's the connection we're going to use between the two repositories.

Now we have our two legs.

What interests us here isn't developing the entire Beta model. It's understanding how Janus uses these two series to construct the regime domains whose topology we want to demonstrate.

In Janus, I'm going to use two quantities to represent the state of our legs: their variances.

Why variances?

Because they measure how spread out their returns are. If the variations stay relatively close to their average, their dispersion is low. If they become much more spread out, it increases.

We can describe this measurement with a first equation:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1} \tag{8}$$

In Janus, the stress variance is calculated using the last 63 observations and annualized using a factor of 252:

$$\begin{aligned} V_t^+ &= 252 \ \mathrm{Var}_{63}(R^+) \\ V_t^- &= 252 \ \mathrm{Var}_{63}(R^-) \end{aligned} \tag{9}$$

Here, this equation is used to measure the recent fluctuations in each of my two legs, then I scale the result to an annual period.

I'm going to place the variance of the negative leg on the horizontal axis and the variance of the positive leg on the vertical axis.

A state now becomes a point with two coordinates:

$$z = (V_-, V_+) \tag{10}$$

This equation simply means: "To locate my state, I use the variances of my two legs."

Why two coordinates? Because I want to study their relationship, not just the dispersion of one isolated leg.

But be careful: variance alone isn't enough.

Imagine a variance of 0.06. Is that a lot? Well, that depends on what the leg has experienced in the past.

If its historical reference is 0.03, the current variance is twice that reference. If the reference is already 0.06, there's no excess at all.

So Janus calculates a historical reference for each leg, which I'll call b_+ and b_-.

In the configuration we're using, this reference is the median of the previous 252 variance observations.

Then it calculates what I call variance stress:

$$\begin{aligned} s_+ &= \max\left(\frac{V_+}{b_+} - 1, 0\right) \\ s_- &= \max\left(\frac{V_-}{b_-} - 1, 0\right) \end{aligned} \tag{11}$$

I measure how much the current variance exceeds its reference. If it doesn't exceed it, I keep zero.

The symbol max simply means "take the larger of the two values."

For example:

$$s_+ = \max\left(\frac{0.06}{0.03} - 1, 0\right) = 1 \tag{12}$$

4 Variance response and the two masses

Source passage: Karim Khem, October 23, 2026.

This result means the variance is 100% above its reference.

Now, look at something important.

Even if I keep the exact same variance, its stress can change when its historical reference changes.

The same point on our map can therefore be interpreted differently when the parameters defining the framework change.

And that's only part of the mechanism.

In Janus, I also measure how the excess returns of the two legs respond to variance. The code estimates a relationship of this form:

$$E = \alpha - mU + \varepsilon \tag{13}$$

Excess return is represented by a base level, minus a sensitivity multiplied by variance, plus the part that the relationship doesn't explain.

Here, E represents excess return, U is the variance used in this estimation, m is the sensitivity, α is the base level, and ε is the residual.

I'm using the letter m so we don't confuse this internal Janus coefficient with the relational Beta object.

When this coefficient is positive, the estimated response decreases as variance increases. When it's negative, the estimated response increases with variance.

Janus retains the admissible parts of these responses to construct two masses:

$$\begin{aligned} f &= \max(m_+, 0) \\ n &= \max(-m_-, 0) \end{aligned} \tag{14}$$

5 Fragile density, antifragile pressure, and balance

Source passage: Karim Khem, October 23, 2026.

Read the first one as: "I keep the positive part of the fragile sensitivity of the positive leg."

And the second: "I keep the positive magnitude of the negative response of the negative leg."

These are the masses used to calculate the model's two densities.

For this first lesson, you don't need to reconstruct the entire estimation process.

The Janus functions already handle that.

What matters to us is understanding how these masses help determine the geometry of our regimes.

And now we arrive at an essential equation.

Janus constructs a density for the fragile component:

$$\rho_+ = \frac{f}{1 + s_+} \tag{15}$$

I take the fragile mass and divide it by one plus the stress of the positive leg.

Notice that stress is in the denominator.

The more it increases, the more this density decreases, assuming the mass remains fixed.

For antifragile pressure, Janus constructs:

$$\rho_- = \frac{n\, s_+}{1 + s_-} \tag{16}$$

I take the mass of the antifragile channel, multiply it by the positive stress that activates it, then divide it by one plus the stress of the other leg.

Now we have two densities that depend on the model's conditions.

And to find out which one dominates, all we have to do is compare them.

In words:

$$\text{Balance} = \text{Fragile density} - \text{Antifragile pressure} \tag{17}$$

Using the model's symbols

$$D = \rho_+ - \rho_- \tag{18}$$

Look how simple this is.

If I take a fragile density of 3 and an antifragile pressure of 2, I get:

$$D = 3 - 2 = +1 \tag{19}$$

The balance is positive.

The fragile component dominates.

Now, if antifragile pressure increases to 4:

$$D = 3 - 4 = -1 \tag{20}$$

6 The strict-dominance domains

Source passage: Karim Khem, October 23, 2026.

The balance is negative.

Antifragile pressure dominates.

We've just found a mathematical rule that distinguishes our two domains.

I'll call F_t the fragile-dominance domain and A_t the antifragile-dominance domain, at a given date t.

$$F_t = \{z : D_t(z) > 0\} \tag{21}$$

$$A_t = \{z : D_t(z) < 0\} \tag{22}$$

"I gather all states whose balance is positive" and "I gather all states whose balance is negative."

Here, the word regime refers to these dominance domains. Janus also has a temporally confirmed state, which we'll discuss after the geometry.

Now we can start drawing the regime domains, because we know exactly how to determine which side each point belongs to.

Look at this first diagram, constructed using the parameters from the Beta-to-Janus calculation.

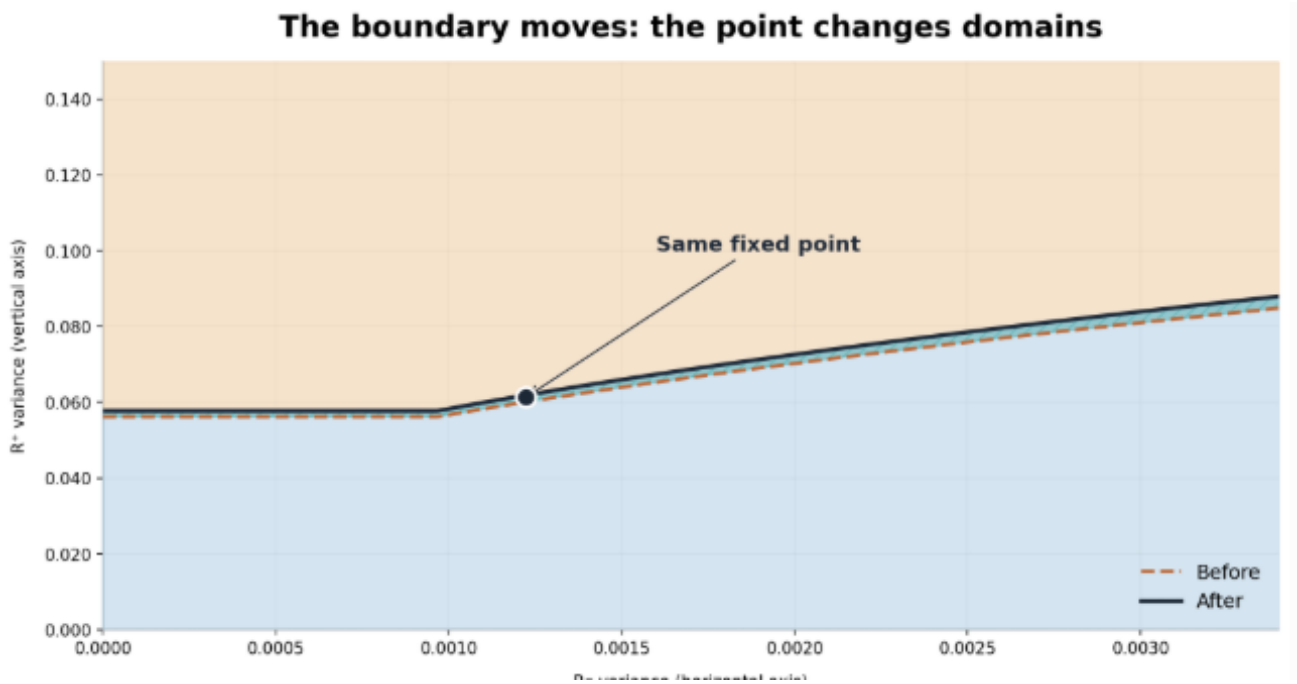


Figure 1: A moving boundary around a fixed state. The blue and orange regions represent fragile and antifragile dominance; dashed and solid curves represent the old and new boundaries. Original figure from the source lesson.

7 Derivation of the equilibrium boundary

Source passage: Karim Khem, October 23, 2026.

Blue represents the fragile domain.

Orange represents the antifragile domain.

The solid line represents the new boundary, and the dashed line represents the old one.

This diagram is a mathematical map of variance states, not an impenetrable price envelope. The axes represent two annualized variances.

Look at the black point.

It stays in exactly the same place, and yet the boundary moves around it. Under the old framework, that point belongs to the antifragile domain. Under the new one, it belongs to the fragile domain.

We're back to the central idea of the Reel:

a point doesn't necessarily have to move to change domains.

The framework around it can evolve too.

But I'm not going to ask you to believe the drawing. We're going to calculate that boundary and prove why it has this particular shape.

First, I need to find the place where the two densities are in equilibrium.

That happens when:

$$D = 0 \tag{23}$$

Since D is the difference between the two densities, this means

$$\rho_+ = \rho_- \tag{24}$$

At the boundary, fragile density is exactly equal to antifragile pressure.

Now I replace these two densities with the equations we've just learned

Since both stress values are nonnegative, both denominators are positive. So instead of keeping the fractions, I'm going to multiply both sides by their denominators straight away.

This gives us:

$$f(1 + s_-) = n\, s_+(1 + s_+) \tag{25}$$

It's exactly the same story, just written in a simpler way using the variables of Janus.

The fragile mass multiplied by one plus negative stress is equal to the antifragile mass multiplied by positive stress and one plus that same positive stress.

And remember, we're still looking for the exact point where our two densities are equal.

That's the boundary between our two regimes.

To find that boundary, I need to isolate positive stress, s_+.

Since both masses are strictly positive, I can divide both sides by n:

$$\frac{f}{n}(1 + s_-) = s_+(1 + s_+) \tag{26}$$

The equation still describes exactly the same equilibrium.

We've simply divided both sides by the same positive quantity to make the next calculation easier.

Then I expand the right-hand side:

$$s_+^2 + s_+ = \frac{f}{n}(1 + s_-) \tag{27}$$

At the boundary, positive stress squared plus that same positive stress must equal the ratio of the two masses multiplied by one plus negative stress.

We've found the equation that determines the critical stress.

To isolate its value, I multiply the equality by 4 and add 1 to both sides:

$$4s_+^2 + 4s_+ + 1 = 1 + 4\frac{f}{n}(1 + s_-) \tag{28}$$

Why?

Because the expression on the left can be written as a square:

$$(2s_+ + 1)^2 = 1 + 4\frac{f}{n}(1 + s_-) \tag{29}$$

I take the square root, then isolate s_+.

Since stress cannot be negative, I keep the nonnegative solution:

$$s_+^\star = \frac{\sqrt{1 + 4\frac{f}{n}(1 + s_-)} - 1}{2} \tag{30}$$

The star simply means "critical value."

We've just demonstrated something important:

for every level of stress on the negative leg, there is a critical level of positive stress at which the two densities are exactly equal.

Below that critical level, the fragile component dominates.

Above it, antifragile pressure dominates.

And now we're going to turn this result into an actual geometric boundary on our variance map.

To do that, I return to an equation you already know:

$$s_+ = \frac{V_+}{b_+} - 1 \tag{31}$$

This relationship is valid at our regular boundary, where positive stress is strictly greater than zero.

I want to find the value of V_+, because that's the variance we placed on the vertical axis.

So I rearrange the equation:

$$V_+ = b_+(1 + s_+) \tag{32}$$

It tells us: "Positive variance is equal to its reference multiplied by one plus its stress."

Now I replace stress with the critical value we just found.

And I get:

$$C_t(V_-) = \frac{b_{+,t}}{2}\left[1 + \sqrt{1 + 4\frac{f_t}{n_t}\max\left(1, \frac{V_-}{b_{-,t}}\right)}\right] \tag{33}$$

8 Interpretation of the boundary equation

Source passage: Karim Khem, October 23, 2026.

There it is. That's our boundary equation.

It might look a little long, but look closely at what it tells us.

On the left, $C_t(V_-)$ represents the height of our boundary, at a particular date and for a particular value of negative variance.

On the right, we find exactly the quantities we've already explained: the two masses and the two variance references.

In other words:

The height of the boundary is determined by the relationships Janus measures between the two legs, and that height changes when the model's parameters change.

You can even understand the deformations simply by reading the equation.

Look at this ratio:

$$\frac{f_t}{n_t} \tag{34}$$

It's simply fragile mass divided by the intensity of the antifragile channel.

When the variance references are held constant, increasing this ratio also increases the critical height.

That means the boundary moves upward on our chart.

Now look at:

$$\max\left(1, \frac{V_-}{b_-}\right) \tag{35}$$

As long as negative variance doesn't exceed its reference, this expression stays equal to 1. So the height of the boundary doesn't yet depend on V_-.

That's why our boundary starts with a horizontal section.

But once negative variance exceeds its reference, the ratio V_-/b_- becomes greater than 1.

The expression starts increasing with that variance, and the boundary curves upward.

So the geometry we're looking at isn't a drawing chosen at random.

IT'S THE SHAPE IMPOSED BY THE EQUATIONS OF JANUS.

Here is that shape using the two successive frameworks of March 16 and 17, 2016, calculated from the Beta legs connected to Janus.

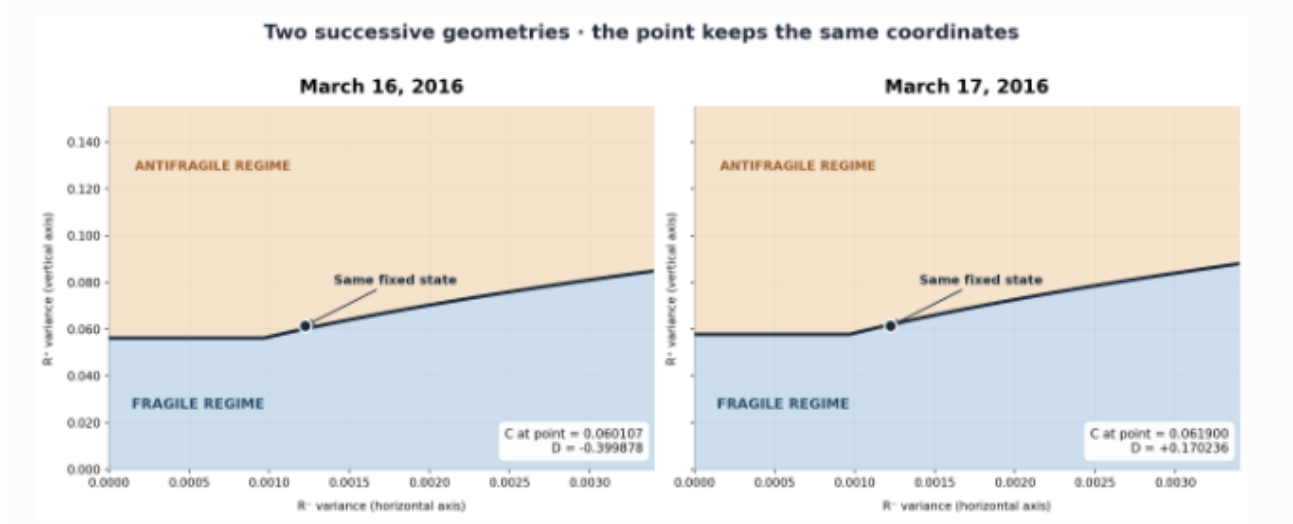


Figure 2: Two successive geometries for March 16 and March 17, 2016. The state keeps the same variance coordinates while the framework and classification change. Original figure from the source lesson.

9 Regime geometry and neighborhoods

Source passage: Karim Khem, October 23, 2026.

Look at the two images.

Blue represents all the states in which the fragile component dominates. Orange represents the states in which antifragile pressure dominates.

The black line separates the two domains.

And now I can write these two regions mathematically.

The fragile domain:

$$F_t = \{(V_-, V_+) : V_+ < C_t(V_-)\} \tag{36}$$

I take every pair of variances whose vertical coordinate is lower than the height of the boundary.

The antifragile domain:

$$A_t = \{(V_-, V_+) : V_+ > C_t(V_-)\} \tag{37}$$

I take every pair of variances located above the boundary.

And finally, the boundary itself:

$$\Gamma_t = \{(V_-, V_+) : V_+ = C_t(V_-)\} \tag{38}$$

Γ, simply gives our boundary a name.

We've now written down the two domains and the boundary separating them.

And that's fundamental for topology.

Why?

Because now I can answer mathematically the questions we asked at the beginning of the lesson.

Does each domain form a single connected region? Can a state move slightly without leaving its domain? Can we connect two states with a path that stays inside? What happens when the boundary deforms?

Let's start with neighborhoods.

Imagine a point inside the fragile domain, sufficiently far from its boundary.

If I change its two variances very slightly, there is a small region around it in which the balance remains positive.

I can call that small region a neighborhood.

To write it mathematically, I take a point z and a small positive number ε.

$$B_\varepsilon(z) = \{z' : \|z' - z\| < \varepsilon\} \tag{39}$$

I gather all the points whose distance from z is smaller than ε.

The symbol $\|\cdot\|$ measures the usual distance between two points on the chart.

Why introduce distance here?

Simply because it lets us describe and draw a neighborhood.

Topology then studies the neighborhood relationships and the properties they define.

If our point lies strictly inside the fragile domain, we can find a neighborhood small enough to stay entirely within that domain.

Mathematically:

$$z \in F_t \implies \exists \varepsilon > 0 : B_\varepsilon(z) \cap X \subset F_t \tag{40}$$

This expression might look complicated, but all it says is:

"If my point lies inside the fragile domain, then there is a small region around it that remains entirely inside the same domain."

X represents the set of possible pairs of variances, while $\subset$ means "is contained in."

Why is this property true?

Because our balance D is a continuous function of the two variances when the parameters defining the framework are fixed and the variance references are strictly positive.

A point whose balance is strictly positive cannot become negative under an arbitrarily small perturbation.

To change its sign, the state must encounter the boundary, or the conditions defining the balance must change.

We've just demonstrated a first property of our domains: every strictly interior point has a neighborhood that preserves its dominance.

But now imagine moving closer to the boundary.

The margin becomes smaller and smaller.

At the boundary, the balance is exactly zero. You can find fragile and antifragile states as close as you like to that separation.

And that's what allows us to demonstrate that, when both masses are positive:

$$\partial F_t = \partial A_t = \Gamma_t \tag{41}$$

The symbol ∂ represents the boundary of a set, in our state space X.

So this equality tells us:

The fragile domain and the antifragile domain share exactly the same boundary.

Now let's move on to connectedness.

Go back to the drop of milk from the Reel.

It could stretch, spread out, change its appearance completely, and still remain one single piece.

Can we demonstrate the same property for each of our two Janus domains?

Yes, in regular frameworks where both masses are strictly positive.

And I'm going to show you a very simple method.

We have a curved boundary whose height changes depending on the horizontal position.

Instead of measuring the point's height directly, I'm going to measure its height relative to the height of the boundary.

I write:

$$u = \frac{V_+}{C_t(V_-)} \tag{42}$$

I take the height of the point and divide it by the height of the boundary at exactly the same horizontal position.

Imagine the boundary has a height of 10.

If your point is at 5:

$$u = 5/10 = 0.5 \tag{43}$$

The point is halfway to the critical height.

If it's at 10:

$$u = 10/10 = 1 \tag{44}$$

It's exactly on the boundary.

And if it's at 15:

$$u = 15/10 = 1.5 \tag{45}$$

It's above the boundary.

Do you see what we've just done?

We've replaced a curved boundary with a reference that is always equal to 1.

Here's what that transformation produces.

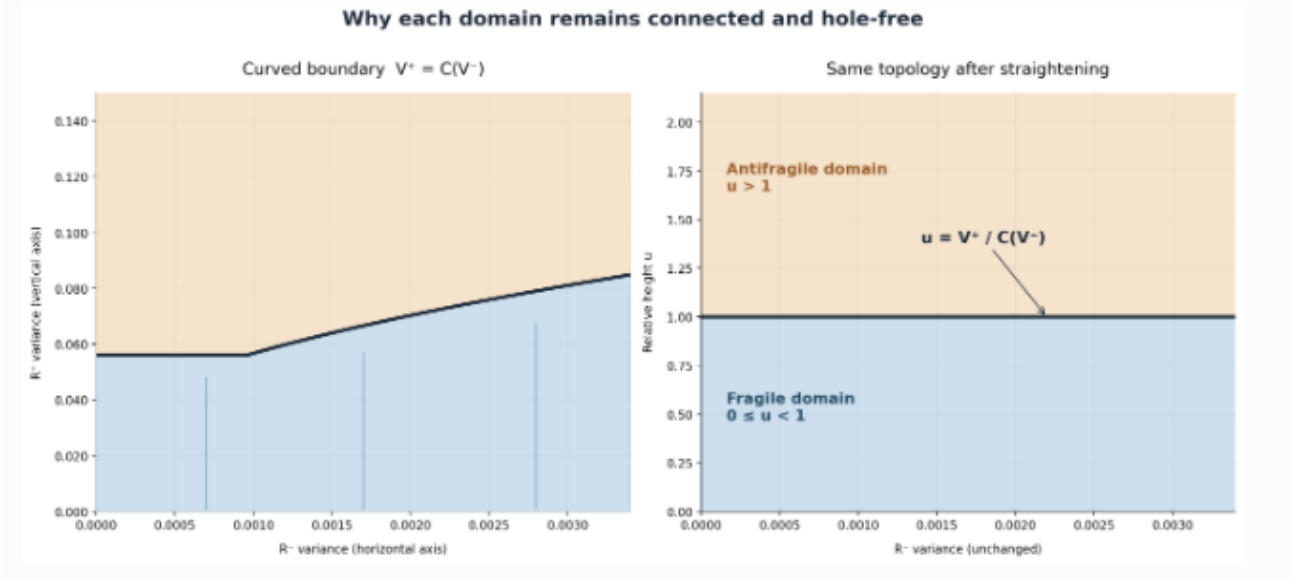


Figure 3: Straightening the boundary using $u = V_+/C_t(V_-)$. The transformed strict-dominance domains lie below and above $u = 1$. Original figure from the source lesson.

On the left, you can see the original geometry of Janus.

On the right, we've straightened the boundary.

Now look at the two domains.

The fragile domain becomes:

$$0 \leq u < 1 \tag{46}$$

And the antifragile domain:

$$u > 1 \tag{47}$$

The boundary becomes:

$$u = 1 \tag{48}$$

Much easier to understand, isn't it?

You now have two regions separated by a horizontal line.

Take any two points in the blue region.

Can you connect them with a straight line without leaving that region?

Yes.

You can draw a segment between them, and every point on that segment remains inside the blue region.

We can demonstrate that with an equation:

$$\gamma(\lambda) = (1 - \lambda)z_1 + \lambda z_2, \qquad 0 \leq \lambda \leq 1 \tag{49}$$

10 Paths, contractibility, and homeomorphism

Source passage: Karim Khem, October 23, 2026.

This equation tells us: "I start at the first point, then progressively move toward the second until I reach it."

When $\lambda = 0$, I'm at the first point.

When $\lambda = 1$, I'm at the second.

Between those two values, I'm moving along the segment.

If both starting heights are below 1, every intermediate weighted combination remains below 1.

The same reasoning works in the antifragile domain, where both heights are greater than 1.

Each domain is therefore path-connected: I can connect any two of its states with a continuous path that remains inside the domain.

And we can go a little further.

A domain can be a single connected piece and still contain a hole. Imagine a ring: you can walk all the way around it, but you can't shrink a loop surrounding its hole into a point without leaving the ring.

Do our two regions contain holes?

Look at our straightened regions.

I can take a loop inside either one and gradually shrink it toward a point chosen inside that same domain.

Mathematically, I can write that deformation as:

$$H(z, \lambda) = (1 - \lambda)z + \lambda z_0 \tag{50}$$

Every point of my loop progressively moves toward the point z_0

Because our straightened regions are convex, these movements remain entirely inside their respective regions.

So we can contract any loop into a point.

This property is called contractibility.

And because we can return from our straightened regions to the original curved regions through a continuous, reversible transformation:

$$V_+ = u\, C_t(V_-) \tag{51}$$

the two representations have the same topological organization.

This is what we call a homeomorphism.

The word sounds complicated, but it describes a simple idea:

"I can continuously and reversibly transform a shape without losing its connections or artificially creating holes."

We've therefore demonstrated that our two regular domains are both path-connected and contractible.

Their geometry can change while these topological properties remain the same.

And now we arrive at exactly what interests me in the Reel: the deformation of the framework itself.

Go back to our boundary equation:

$$C_t(V_-) = \frac{b_{+,t}}{2}\left[1 + \sqrt{1 + 4\frac{f_t}{n_t}\max\left(1, \frac{V_-}{b_{-,t}}\right)}\right] \tag{52}$$

11 A changing framework: March 16-17, 2016

Source passage: Karim Khem, October 23, 2026.

Notice that little t?

It represents time.

The masses and references defining the framework are recalculated using the available historical data.

So our boundary can change from one date to another.

I can write that more simply:

$$C_{t_1}(V_-) \neq C_{t_2}(V_-) \tag{53}$$

At the same horizontal position, the boundary can have a different height at two different dates.

And if that boundary moves far enough, it can end up on the other side of a state that we keep fixed.

We've actually found this mechanism in the data from the two repositories.

Let's take March 16, 2016.

The calculated parameters include:

$$\begin{aligned} f_{16} &\approx 10.292917 \\ n_{16} &\approx 10.115890 \end{aligned} \tag{54}$$

The positive variance reference is approximately:

$$b_{+,16} = 0.0345594225 \tag{55}$$

And the negative variance reference is:

$$b_{-,16} = 0.0009679545 \tag{56}$$

I take the state observed on that date:

$$z = (0.0012231272,\ 0.0613259825) \tag{57}$$

The first coordinate is negative-leg variance. The second is positive-leg variance.

By replacing the parameters in our boundary equation, I get the following critical height at that horizontal position:

$$C_{16}(V_-) \approx 0.060107381 \tag{58}$$

The height of the point is 0.0613259825.

That's higher than the boundary:

$$0.0613259825 > 0.060107381 \tag{59}$$

So the point belongs to the antifragile domain.

To verify this another way, I calculate the two Janus densities at the same point.

They are approximately:

$$\begin{aligned} \rho_{+,16} &= 5.800433 \\ \rho_{-,16} &= 6.200311 \end{aligned} \tag{60}$$

The balance becomes:

$$D_{16} = 5.800433 - 6.200311 \approx -0.399878 \tag{61}$$

It's negative.

We get exactly the same dominance.

Now watch what I'm going to do.

I keep both coordinates of my point exactly the same. I don't change them.

I simply replace the March 16 framework with the parameters calculated for March 17.

The masses become approximately:

$$\begin{aligned} f_{17} &= 9.996732 \\ n_{17} &= 8.985637 \end{aligned} \tag{62}$$

The positive variance reference becomes:

$$b_{+,17} = 0.0346433700 \tag{63}$$

The negative reference remains practically unchanged in this comparison.

And when I recalculate the boundary using these new parameters, I get:

$$C_{17}(V_-) \approx 0.061900145 \tag{64}$$

Now look at this relationship:

$$0.060107381 < 0.0613259825 < 0.061900145 \tag{65}$$

It tells us that the height of our fixed point lies between the old boundary and the new boundary.

Under the first framework, the point is above the boundary.

Under the second, it's below.

So its new classification is fragile.

And if I calculate the two densities directly under the new framework:

$$\begin{aligned} \rho_{+,17} &\approx 5.647207 \\ \rho_{-,17} &\approx 5.476971 \end{aligned} \tag{66}$$

The balance becomes:

$$D_{17} = 5.647207 - 5.476971 \approx +0.170236 \tag{67}$$

It's positive.

Now look at the enlarged chart.

The dashed line represents the old boundary. The solid line represents the new one.

The black point is exactly the same in this comparison.

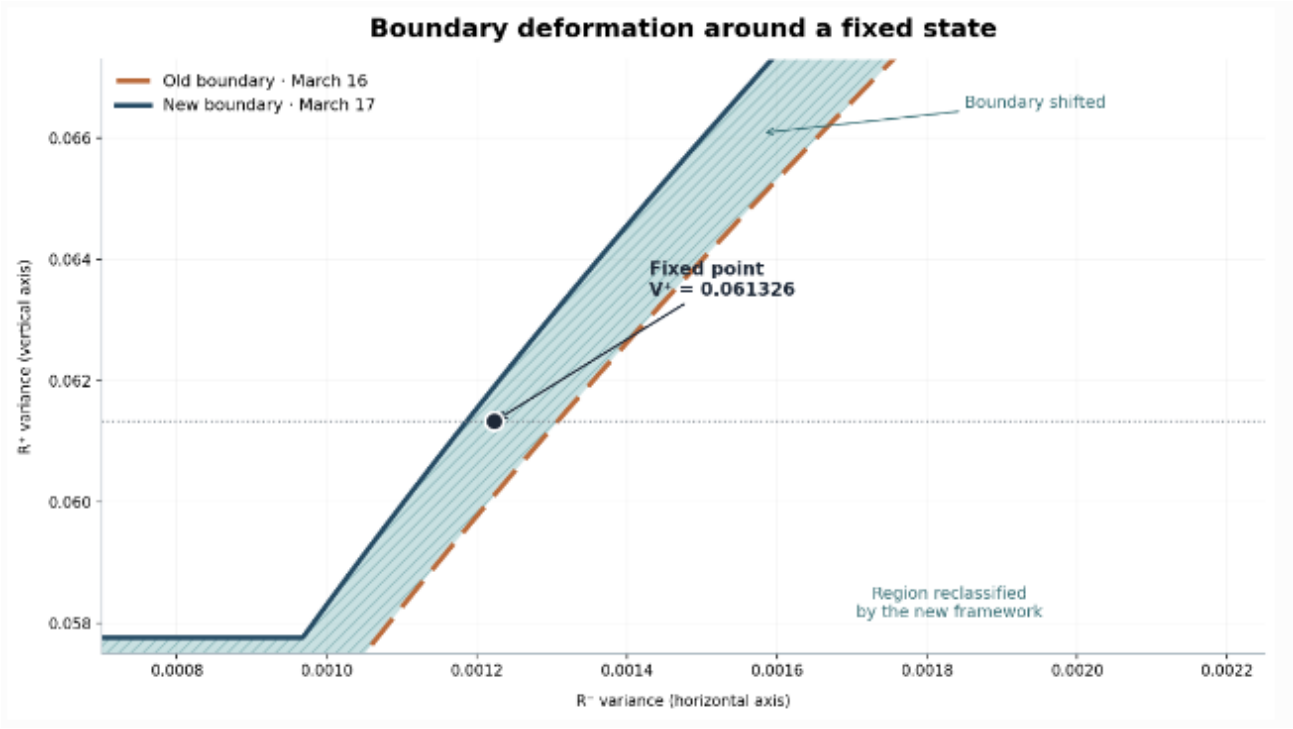


Figure 4: Boundary deformation around a fixed state. The hatched region contains states between the old and new boundaries and corresponds to the reclassification set Z. Original figure from the source lesson.

The hatched region represents all the states that change domains because of this modification to the framework.

We can even write that region mathematically:

$$Z = \{(V_-, V_+) : C_{16}(V_-) < V_+ < C_{17}(V_-)\} \tag{68}$$

12 Reclassification through boundary deformation

Source passage: Karim Khem, October 23, 2026.

I gather all the states located above the old boundary but below the new one.

So it's not just the coordinates of our particular point that matter.

An entire region of the state space changes domains when we update the framework.

And in our calculation, we can distinguish the two effects.

If I change only the observed variances while keeping the March 16 framework, the balance remains negative in this episode.

But if I keep the old variances and change only the parameters defining the framework, the balance becomes positive.

So we've isolated the effect of the boundary's deformation.

This is a mathematical comparison using parameters derived from historical data, not an observation in which market parameters physically changed independently of the variances used to estimate them.

That distinction is precisely what allows us to understand the mechanism.

We've demonstrated that the geometry of the domains can evolve enough to reclassify a fixed state.

And now let's return to topology.

During this deformation, the fragile domain still exists, and so does the antifragile domain.

Each remains a single connected region.

Their shared boundary changes its shape and position, but their topological type is preserved.

It's comparable to our drop of milk being stretched without being split apart.

But are there also situations where more than just the geometry changes?

Yes. And we're going to demonstrate that directly using an equation from Janus.

Go back to antifragile pressure:

$$\rho_- = \frac{n\, s_+}{1 + s_-} \tag{69}$$

13 The appearance of a second domain

Source passage: Karim Khem, October 23, 2026.

Now ask yourself a very simple question.

What happens if $n = 0$?

The answer is right there in the equation:

$$\rho_- = \frac{0 \times s_+}{1 + s_-} = 0 \tag{70}$$

Antifragile pressure is zero everywhere in our stress space.

And if fragile mass remains strictly positive, then:

$$D = \frac{f}{1 + s_+} > 0 \tag{71}$$

The balance remains strictly positive for every finite level of stress.

In other words, the fragile domain occupies the entire state space, and the antifragile domain doesn't exist under this framework.

We can write:

$$F = X, \qquad A = \varnothing \tag{72}$$

The symbol X represents the set of possible states. The symbol $\varnothing$ simply means "empty set."

Now imagine that n becomes positive while f remains positive.

Antifragile pressure can then become dominant once positive stress exceeds a certain level.

A finite boundary appears, separating the two domains.

So we move from a configuration in which only one dominance domain is present to a configuration in which two domains are present.

And we've also found this in the Beta data connected to Janus.

On April 9, 2020, the calculated masses were:

$$\begin{aligned} f_9 &\approx 4.835370 \\ n_9 &= 0 \end{aligned} \tag{73}$$

The antifragile domain was absent.

On April 10:

$$\begin{aligned} f_{10} &\approx 4.713758 \\ n_{10} &\approx 0.254082 \end{aligned} \tag{74}$$

Both masses are now positive.

Here's the transformation.

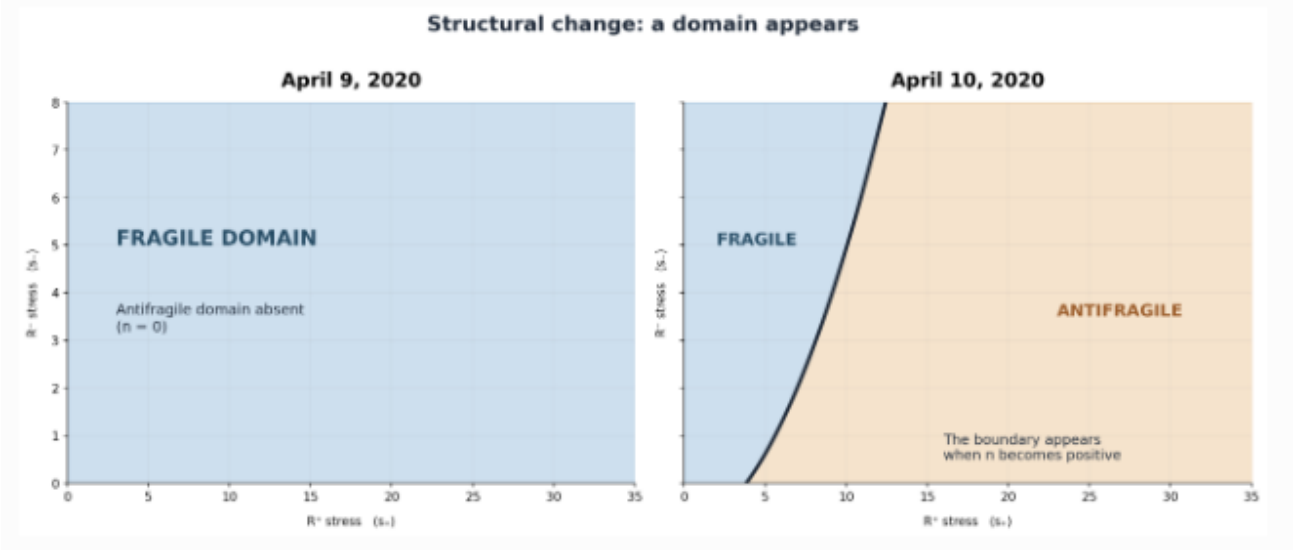


Figure 5: The appearance of an antifragile domain between April 9 and April 10, 2020. Here the axes are the two stress coordinates. The absence of the antifragile domain on April 9 follows from $n = 0$, rather than from the limits of the plotted window. Original figure from the source lesson.

For this particular drawing, I chose stress as our coordinates: positive stress on the horizontal axis and negative stress on the vertical axis.

This allows us to show the role of n directly, without bringing the variance references into the drawing.

Look at the image on the left.

The fragile domain occupies the entire region represented.

The antifragile domain is absent because its pressure is zero.

In the image on the right, a boundary has appeared.

Both domains now exist.

It's important to understand that you cannot conclude a domain is empty simply because you don't see it within the window of a chart.

Here, its absence is demonstrated by the equation with $n = 0$.

When $n > 0$, the antifragile domain exists beyond a finite boundary, even if that boundary is too high to appear within a small plotting window.

We can now count the connected components present in our partition of strict-dominance domains.

I write:

$$S = F \cup A \tag{75}$$

14 Connected components and topological change

Source passage: Karim Khem, October 23, 2026.

I combine the two dominance domains into one set.

When $n = 0$ and $f > 0$, only one domain is present.

So S has one connected component.

When $f > 0$ and $n > 0$, we have two domains separated by their boundary, which does not belong to S.

The set S then has two connected components.

$$\text{Number of connected components:} \quad 1 \longrightarrow 2 \tag{76}$$

15 Instantaneous dominance and temporal confirmation

Source passage: Karim Khem, October 23, 2026.

The organization of the partition has changed from one dominance domain to two distinct domains.

This time, we're not dealing with geometric deformation alone. The number of connected components of the partition has changed.

So we've found two different mathematical phenomena in Janus.

In the first, both domains are present and their boundary deforms. Their geometry changes, but their topological properties remain the same.

In the second, a previously absent domain appears. The organization of the partition itself changes.

And this is where we can speak precisely about a topological change in the regime partition.

But there's one last question.

Does crossing to the other side of the boundary mean Janus has immediately confirmed a regime change?

No, because Janus distinguishes between instantaneous dominance and its temporally confirmed state.

I'll call that state G_t.

The balance D_t answers one question: "Which component dominates right now?"

The state G_t answers another: "Have the conditions required to confirm the new regime been met?"

In the Janus code, entering the confirmed pressure state requires the balance to remain negative for a duration calculated by the model.

That duration is written as:

$$\tau_t = \text{round}\left[\text{clip}\left(\frac{1}{\sqrt{\max(|D_t|, 10^{-6})}}, 3, 30\right)\right] \tag{77}$$

Let me translate that.

First, I take the absolute value of the balance, meaning its magnitude regardless of whether it's positive or negative.

If that magnitude is large, the quantity

$$\frac{1}{\sqrt{|D_t|}} \tag{78}$$

is small. If the magnitude is small, this quantity increases.

The clip function then limits the result to between 3 and 30 observations, and round rounds it to an integer.

So the complete equation tells us:

"The model adjusts the confirmation period according to the strength of the imbalance between the two densities, while imposing a minimum and maximum duration."

When Janus is waiting to enter the confirmed pressure state, it counts consecutive observations where the balance is negative.

Confirmation occurs when that counter reaches the calculated duration.

To exit that state, its default configuration uses an acceleration signal from the positive leg. So that's a different rule from simply reversing the sign of the balance.

In the April 2020 Beta-to-Janus example, the new domain appears under the April 10 framework, and the temporal Janus state then confirms its transition on April 14.

That distinction matters.

The boundary determines geometric dominance. The temporal rule determines operational confirmation.

We now have a much more complete mathematical representation of what I explained in the Reel.

Price movements feed the variations of the two legs.

Those variations feed the variances, historical references, and estimated responses.

Those quantities determine the two Janus densities.

The densities define the domains and their boundary.

And when the parameters change, the geometry of those domains can change with them.

Topology becomes useful because it allows us to say precisely what is preserved and what is not: neighborhoods, paths, connectedness, the presence of domains, and their separation.

Topology doesn't automatically tell us tomorrow's price. It allows us to construct and study the organization of the states in which our model can evolve.

And that's exactly what we wanted to understand.

16 Computational reconstruction

Source passage: Karim Khem, October 23, 2026.

Now you need to be able to reproduce this demonstration yourself.

For your first exercise, I'm going to ask you to reconstruct the topology of the two Janus regimes using the Beta data and the equations we've just explained.

You'll start with the two series R^+ and R^-, found in:

`data/beta_spread_pack_daily_sanitized.csv`

You'll transform them into two positive indices, just as we did in this lesson, then use the Janus functions to obtain the masses, historical references, and necessary diagnostics.

You don't have to reinvent their estimation methods.

The goal is to understand how the quantities produced by the algorithm become a geometry of market regimes.

Then you return to our equations:

$$\begin{aligned} s_+ &= \max\left(\frac{V_+}{b_+} - 1, 0\right) \\ s_- &= \max\left(\frac{V_-}{b_-} - 1, 0\right) \end{aligned} \tag{79}$$

Then:

$$\begin{aligned} \rho_+ &= \frac{f}{1 + s_+} \\ \rho_- &= \frac{n\, s_+}{1 + s_-} \end{aligned} \tag{80}$$

And finally:

$$D = \rho_+ - \rho_- \tag{81}$$

17 Python skeleton and reconstruction tasks

Source passage: Karim Khem, October 23, 2026.

These are the equations you're going to translate into a few lines of Python.

Here's the skeleton:

```
import numpy as np

def calculate_topology(V_plus, V_minus, f, n, b_plus, b_minus):

    stress_plus = ...
    stress_minus = ...

    rho_plus = ...
    rho_minus = ...

    balance = ...

    fragile = balance > 0
    antifragile = balance < 0

    return balance, fragile, antifragile
```

All the answers are already in the lesson.

Next, you calculate this balance across a grid of variance pairs, coloring fragile states in blue and antifragile states in orange.

You draw the boundary wherever the two densities are equal, when both masses are positive.

I want you to reconstruct the map for March 16, 2016, and then the one for March 17.

You keep exactly the same point on both drawings.

And you should recover this result:

$$D_{16} < 0 \qquad \text{and} \qquad D_{17} > 0 \tag{82}$$

In this fixed-point comparison, dominance shifts from antifragile to fragile.

But most importantly, I want you to explain why.

You need to be able to show that the parameters defining the framework changed, causing the boundary to move around the point.

Then I want you to straighten that boundary using:

$$u = \frac{V_+}{C(V_-)} \tag{83}$$

You should recover the two regions separated by $u = 1$ and explain why they are connected.

Finally, go back to the case where $n = 0$.

I want you to demonstrate, directly from the equation for ρ_-, why the antifragile domain is absent when $f > 0$.

Then explain what changes when both masses become positive.

You've already got all the answers in the lesson.

All that's left is to translate them into calculations and reproduce the diagrams.

And that's the principle behind this series of lessons: I'm going to explain the mathematical properties we need, show you how they work inside the model, then ask you to reconstruct them yourself in code.

For this first stage, we're stopping at the topology of the two regimes and their transitions. The other mechanisms of the algorithm will come later.

You now have what you need to understand why the boundary isn't just a line added to a chart.

It's calculated from the model's equations. Its shape depends on the estimated conditions. Its deformation changes the domains, and certain changes even alter the topological organization of their partition.

PRICES EVOLVE, MARKET CONDITIONS EVOLVE, AND THE DOMAINS OF YOUR MODEL CAN EVOLVE WITH THEM. THAT'S WHY YOU CAN'T JUST STUDY WHAT MOVES INSIDE THE FRAMEWORK. YOU ALSO NEED TO UNDERSTAND HOW THE FRAMEWORK ITSELF TRANSFORMS.

18 Discussion and conclusions

The construction separates three questions. First, the instantaneous balance determines which density dominates at a state. Second, the boundary geometry describes how the classification changes across variance states and across parameter settings. Third, the temporal rule determines when the algorithm confirms an operational transition. These objects should remain distinct.

For positive masses and positive references, the explicit boundary is continuous and strictly positive. Its normalized coordinate places the two strict-dominance domains in convex regions separated by $u = 1$. Each domain is therefore path-connected and contractible, even as its boundary bends or moves. By contrast, the transition from $n = 0$ to $n > 0$ with $f > 0$ changes the number of connected components in the union of strict-dominance domains, because the equality boundary is excluded from that union. The ambient state space X itself has not been split or changed.

The March 2016 example is a fixed-state comparison across two estimated frameworks. It illustrates the mathematical effect of changing parameters; it does not establish an intervention that changes those parameters independently of the data used to estimate them. The April 2020 example separates the emergence of a geometric domain from the later confirmation of an operational state. Both examples are reproduced as source-reported illustrations.

The algebra establishes classification and topological properties under the stated assumptions. It does

not, by itself, establish predictive accuracy, profitability, robustness to estimation error, or out-of-sample performance. Those questions require separately specified data, estimation procedures, execution assumptions, and empirical validation. Within that scope, the central conclusion is precise: a market state may be reclassified either because it moves within a fixed framework or because the framework's boundary changes around it.

References

[1] Karim Khem. The topology of market regimes. KHEM KAPITAL SCHOOL channel lesson, October 22-23, 2026. Archived November 1, 2026. https://discord.com/channels/1199900098516226139/1562940477504753775.

[2] karimkhemkapital. Janus anti-fragility strategy. Repository linked in the source lesson. https://github.com/karimkhemkapital/janus-anti-fragility-strategy.

[3] karimkhemkapital. Beta and trading environment. Repository linked in the source lesson. https://github.com/karimkhemkapital/Beta-and-trading-environement.

[4] Karim Khem. Instagram reel linked as prerequisite material in the source lesson. https://www.instagram.com/reel/De3GDaxzmaq/.

A Source fidelity and transcription notes

This edition draws on the saved 181-message channel archive, containing 83 mathematical attachments and five diagrams. Promotional posts, the pinned-message notice, reaction controls, and avatars are not part of the research-style reading text; they remain available in the complete channel archive. The substantive mathematical exposition and reconstruction skeleton are retained.

Two source renderings have been normalized. In attachment 62, the comparison glyph is malformed; the typeset comparison uses $>$, consistent with both displayed numbers and the adjacent statement that the point is above the boundary. In attachment 83, the code-like reciprocal square-root expression is typeset as $1/\sqrt{|D_t|}$. The full confirmation formula retains the 10^{-6} stabilization term. No reported decimal values have been altered.

The accompanying `math-transcription.json` maps every source attachment to its native LaTeX expression or retained figure and records these two normalization notes. The original attachment images are included in the source bundle. The source lesson's code skeleton deliberately contains ellipses: it is an exercise, not a completed implementation.

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